



PARUL UNIVERSITY

Faculty Of Engineering & Technology

Department of Applied Sciences & Humanities

1st year B.Tech Programme (All branches)

Mathematics-II (Subject Code :303191151)

UNIT-6 Multivariable Calculus (Integration)

Double Integration as Iterated Integration:

Double integration is evaluated as iterated integration,

e.g. (i) $\int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy dx,$

We first integrate it with respect to y and then we integrate with respect to x.

$$\int_{x=a}^{x=b} \int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy dx = \int_{x=a}^{x=b} \left(\int_{y=g_1(x)}^{y=g_2(x)} f(x, y) dy \right) dx$$

(ii) $\int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} f(x, y) dx dy$

We first integrate it with respect to x and then with respect to y

$$\int_{y=c}^{y=d} \int_{x=h_1(y)}^{x=h_2(y)} f(x, y) dx dy = \int_{y=c}^{y=d} \left(\int_{x=h_1(y)}^{x=h_2(y)} f(x, y) dx \right) dy$$

Evaluate the following double integrations:

Example: 1 Evaluate $\int_1^2 \int_1^2 \frac{1}{xy} dx dy.$

$$= \int_1^2 \frac{1}{y} \left(\int_1^2 \frac{1}{x} dx \right) dy$$

$$= \int_1^2 \frac{1}{y} [\log x]_1^2 dy$$

$$= \log 2 [\log y]_1^2$$

$$= \log 2 \times \log 2$$

$$= (\log 2)^2$$

Example:2 Evaluate $\int_0^{\pi/2} \int_1^2 y \sin xy dx dy.$

$$= \int_0^{\pi/2} y [\cos xy]_1^2 \frac{1}{y} dy$$

$$= \int_0^{\pi/2} [-\cos 2y + \cos y]_1^2 dy$$

$$= - \int_0^{\pi/2} \cos 2y dy + \int_0^{\pi/2} \cos y dy$$

$$= \left[-\frac{1}{2} \sin 2y + \sin y \right]_0^{\pi/2}$$

$$= 1$$

Examples:

1) $\int_1^2 \int_3^4 (xy + e^y) dy dx$

2) $\int_0^1 \int_0^{x^2} (e^{y/x}) dy dx$

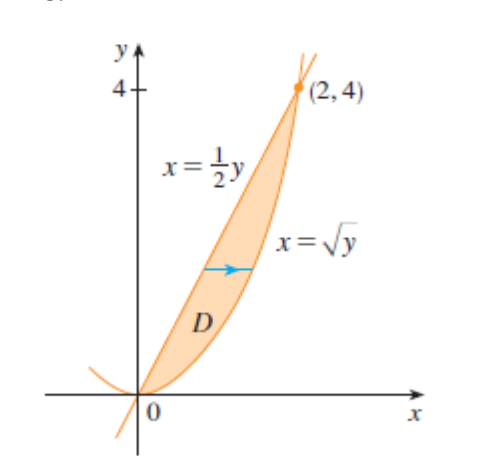
3) $\int_0^1 \int_y^{y^2+1} x^2 y dx dy$

4) $\int_1^3 \int_0^1 (1 + 4xy) dx dy$

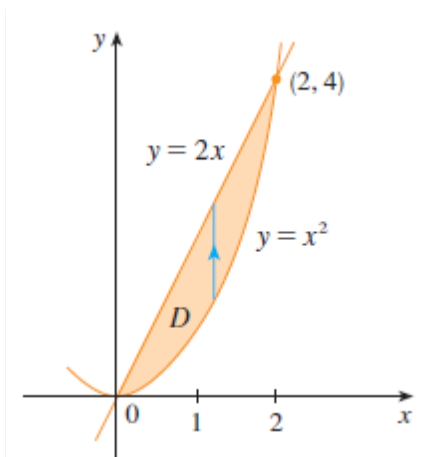
5) $\int_0^{\pi/2} \int_0^2 x \sin y dx dy$

Example 1: Evaluate the $\iint_R (x^2 + y^2) dA$, where R is the region bounded by the line $y = 2x$ and the parabola $y = x^2$.

Ans:



$$\begin{aligned} I &= \iint_R (x^2 + y^2) dA \\ &= \int_0^4 \int_{\frac{y}{2}}^{\sqrt{y}} (x^2 + y^2) dx dy \\ &= \int_0^4 \left(\frac{x^3}{3} + xy^2 \right) \Big|_{\frac{y}{2}}^{\sqrt{y}} dy \\ &= \int_0^4 \left(\frac{y^{\frac{3}{2}}}{3} + y^{\frac{5}{2}} - \frac{13y^3}{24} \right) dy \\ &= \left(\frac{2y^{\frac{5}{2}}}{15} + \frac{2y^{\frac{7}{2}}}{7} - \frac{13y^4}{96} \right) \Big|_0^4 = \frac{216}{35} \end{aligned}$$



$$\begin{aligned} \text{OR} \\ \iint_R (x^2 + y^2) dA &= \int_0^2 \int_{2x}^{x^2} (x^2 + y^2) dy dx \\ &= \int_0^2 \left(x^2 y + \frac{y^3}{3} \right) \Big|_{2x}^{x^2} dx \\ &= \int_0^2 \left(x^4 + \frac{x^6}{3} - \frac{14x^3}{3} \right) dx \\ &= \left(\frac{x^5}{5} + \frac{x^7}{21} - \frac{7x^4}{6} \right) \Big|_0^2 = -\frac{216}{35} \end{aligned}$$

Example 2: Evaluate $\iint_R \sin(y^2) dA$, where R is the region bounded by the lines $y = x$, $y = \pi$, $x = 0$.

$$\int_0^\pi \int_0^y \sin(y^2) dx dy$$

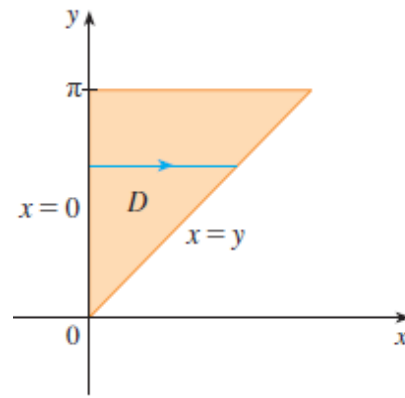
$$= \int_0^\pi y \sin(y^2) dy$$

Suppose $y^2 = t$

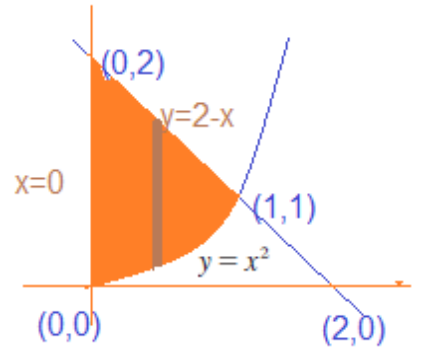
$$\therefore y dy = \frac{dt}{2}$$

$$= \int_0^{\pi^2} \sin t \frac{dt}{2}$$

$$= \left(-\frac{\cos t}{2} \right)_0^{\pi^2} = \frac{1 - \cos(\pi^2)}{2}$$



Example 3: Compute the double integral of the function $f(x, y) = 6 - x + 2y$ over the region bounded by the curves $y = x^2$ and $y = 2 - x$ in the x - y plane.



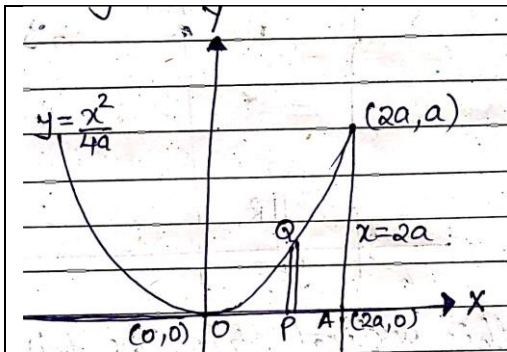
$$= \int_0^1 \int_{x^2}^{2-x} (6 - x + 2y) dy dx$$

$$= \int_0^1 (6y - xy + y^2) \Big|_{x^2}^{2-x} dx$$

$$= \int_0^1 (16 - 12x - 4x^2 + x^3 - x^4) dx$$

$$= \frac{523}{60}$$

Example:4 Evaluate $\iint_R xy dA$, where R is the region bounded x -axis, the ordinates $x=2a$ and the curve $x^2=4ay$.



Limits of y are $y = 0$ to $y = x^2/4a$ and the limits of x

are $x=0$ and $x=2a$ $\iint_R xy dA = \int_0^{2a} \int_0^{x^2/4a} xy dy dx$

$$= \int_0^{2a} [xy^2]_0^{x^2/4a} dx$$

$$= \int_0^{2a} \frac{x^5}{2 \times 16a^2} dx$$

$$= \frac{1}{32a^2} \left[\frac{x^6}{6} \right]_0^{2a}$$

$$= \frac{a^4}{3}$$

Examples:

Example: Evaluate $\iint_R \frac{\sin x}{x} dA$, where R is a triangle in the xy -plane bounded by the x -axis, the lines $y=x$ and $x=1$.

Example: Evaluate $\iint_R (x+y) dA$, where R is bounded by the by $x=0$, $x=2$, $y=x$, $y=x+2$.

Example: Evaluate $\int_0^1 \int_0^x (x^2 + y^2) dA$, where dA indicates small area in xy -plane.

Change of Order:

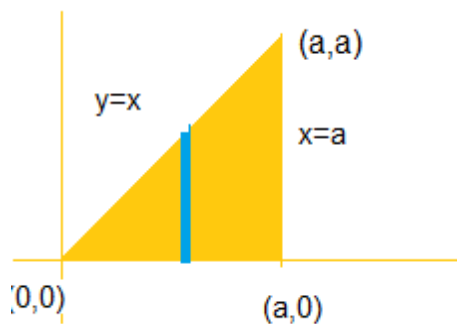
The change of order often makes the evaluation of double integration easier e.g. in example 2 in previous section,

$\int_0^\pi \int_x^\pi \sin(y^2) dy dx$ will be evaluated on reversing the order of integration.

In double integral with constant limits, the order of integration is immaterial provided the limits of integration are changed accordingly. But in case of double integral with variable limits, the limits of the integration changes with the change of order of integration. The new limits are obtained by sketching the region of integration. Sometime in changing the order of integration, it is required to split up the region of integration, and the given integral is expressed as the sum of number of double integrals with the changed limits.

Example 1

Change the order of integration in $\int_0^a \int_y^a \frac{x dx dy}{x^2 + y^2}$, and evaluate the same

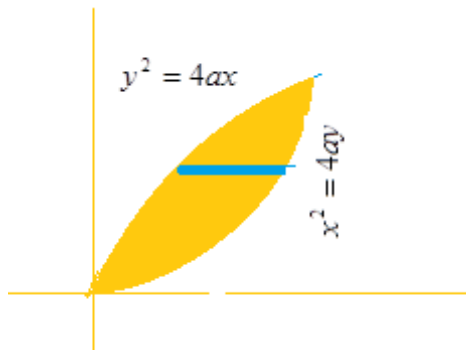


From the limits of integration, it is clear that the region of integration is bounded by $y=x$, $x=a$ and $y=0$. Thus, the region of integration as shown in figure. Draw vertical strip. So new limits of integration are
Limits of x : from $y=0$ to $y=x$
Limits of y : from $x=0$ to $x=a$

$$\begin{aligned} & \int_0^a \int_y^a \frac{x dy dx}{x^2 + y^2} \\ &= \int_0^a x \left(\frac{1}{x} \tan^{-1} \left(\frac{y}{x} \right) \right)_0^x dx \\ &= \int_0^a \frac{\pi}{4} dx \\ &= \frac{\pi}{4} (x)_0^a \\ &= \frac{\pi a}{4} \end{aligned}$$

Example 2:

Change the order of integration in the following integral and evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{az}} dy dx$



After changing the order, new limits are

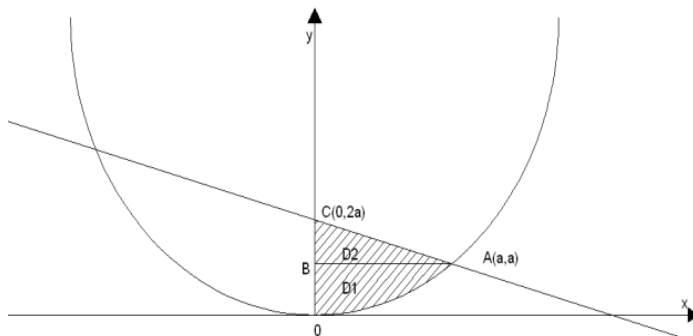
From $x = \frac{y^2}{4a}$ to $x = 2\sqrt{ay}$

From $y = 0$ to $y = 4a$

$$\begin{aligned} \int_0^{4a} \int_{\frac{y^2}{4a}}^{2\sqrt{ay}} dx dy \\ &= \int_0^{4a} (x)_{\frac{y^2}{4a}}^{2\sqrt{ay}} dy \\ &= \int_0^{4a} \left(2\sqrt{ay} - \frac{y^2}{4a} \right) dy \\ &= \left(\frac{4\sqrt{a}y^{\frac{3}{2}}}{3} - \frac{y^3}{12a} \right)_0^{4a} \\ &= \frac{16a^2}{3} \end{aligned}$$

Example 3 Change the order of integration $\int_0^a \int_{\frac{x^2}{a}}^{2a-x} xy dy dx$ and hence evaluate it .

Sol: Given limits $y = \frac{x^2}{a}$ to $y = 2a - x$ and $x = 0$ to $x = a$.



Take horizontal strip to change of integration .

Region is divided in to two sub regions OAB & ABC .

In Region OAB

x varies from $x = 0$ to $x = \sqrt{ay}$,
 y varies from $y = 0$ to $y = a$.

In Region ABC

x varies from $x = 0$ to $x = 2a - y$,
 y varies from $y = a$ to $y = 2a$.

$$\begin{aligned} \int_0^a \int_{\frac{x^2}{a}}^{2a-x} xy dy dx &= \int_{y=0}^a \int_{x=0}^{\sqrt{ay}} xy dx dy + \int_{y=a}^{2a} \int_{x=0}^{2a-y} xy dx dy \\ &= \int_{y=0}^a y \left[\frac{x^2}{2} \right]_0^{\sqrt{ay}} dy + \int_{y=a}^{2a} y \left[\frac{x^2}{2} \right]_0^{2a-y} dy \\ &= \int_{y=0}^a a \frac{y^2}{2} dy + \int_{y=a}^{2a} y \frac{(2a-y)^2}{2} dy \\ &= \frac{a}{2} \left[\frac{y^3}{3} \right]_0^a + \frac{1}{2} \int_{y=a}^{2a} (4a^2y - 4ay^2 + y^3) dy \\ &= \frac{a^4}{6} + \frac{1}{2} \left[2a^2y^2 - \frac{4}{3}ay^3 + \frac{y^4}{4} \right]_a^{2a} \end{aligned}$$

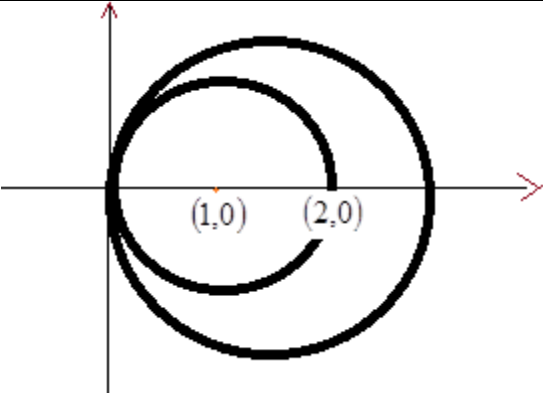
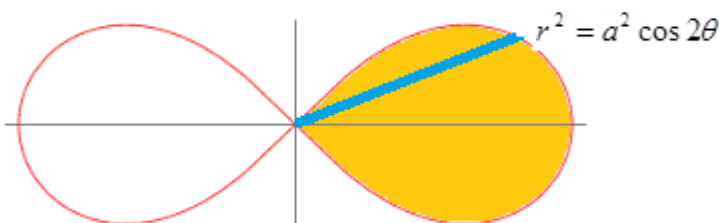
$$= \frac{3a^4}{8}$$

Examples:

1. Evaluate $\int_0^2 \int_0^{4-x^2} \frac{xe^{2y}}{4-y} dy dx$ by changing the order of integration. Ans: $\frac{1}{4}e^8 - 1$.
2. Evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} dx dy$ by changing the order of integration. Ans: $e - 1$.

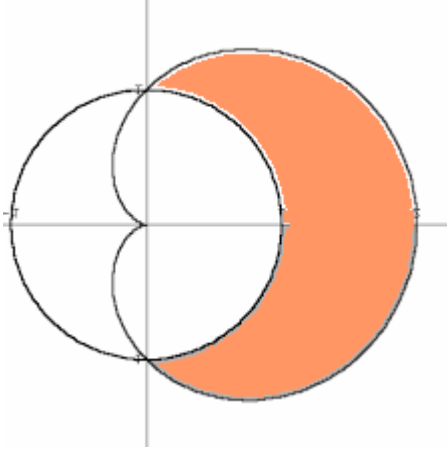
Evaluation of Double integral in Polar Co-ordinates (Applications)

To evaluate the double integral $\iint_R f(r, \theta) dr d\theta$ over the region R bounded by the curves $r = a$, $r = b$ and the straight lines $\theta = \alpha$ and $\theta = \beta$. First integrate with respect to r and θ .

Example 1:	Evaluate $\iint_R r^3 dr d\theta$, over the area bounded between the circles $r = 2\cos\theta$ and $r = 4\cos\theta$.
	$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{2\cos\theta}^{4\cos\theta} r^3 dr d\theta$ $= 2 \int_0^{\frac{\pi}{2}} \int_{2\cos\theta}^{4\cos\theta} r^3 dr d\theta$ $= 2 \int_0^{\frac{\pi}{2}} \left(\frac{r^4}{4} \right)_{2\cos\theta}^{4\cos\theta} d\theta$ $= 120 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$ $= 120 \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$ $= \frac{45\pi}{2}$
Example 2:	Evaluate $\iint_R \frac{r dr d\theta}{\sqrt{a^2 + r^2}}$ over one loop of lemniscates $r^2 = a^2 \cos 2\theta$.
	$\iint_R \frac{r dr d\theta}{\sqrt{a^2 + r^2}}$ $= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \int_0^{a\sqrt{\cos 2\theta}} \frac{r dr d\theta}{\sqrt{a^2 + r^2}}$

$$\begin{aligned}
&= 2 \int_0^{\frac{\pi}{4}} \int_0^{a\sqrt{\cos 2\theta}} \frac{r dr d\theta}{\sqrt{a^2 + r^2}} \\
&= 2 \int_0^{\frac{\pi}{4}} \left(2\sqrt{a^2 + r^2} \right)_0^{a\sqrt{\cos 2\theta}} \\
&= 4 \int_0^{\frac{\pi}{4}} \left(\sqrt{a^2 + a^2 \cos 2\theta} - a \right) d\theta \\
&= 4a \int_0^{\frac{\pi}{4}} (2\cos\theta - 1) d\theta \\
&= 4a \left(2\sin\theta - \theta \right)_0^{\frac{\pi}{4}} \\
&= 4a \left(\sqrt{2} - \frac{\pi}{4} \right)
\end{aligned}$$

Example 3: Find the area of the region that lies inside the cardioid $r = 1 + \cos\theta$ and the circle $r = 1$.



$$\begin{aligned}
&\iint_R r dr d\theta \\
&= 2 \int_0^{\frac{\pi}{2}} \int_1^{1+\cos\theta} r dr d\theta \\
&= 2 \int_0^{\frac{\pi}{2}} \left(\frac{r^2}{2} \right)_1^{1+\cos\theta} d\theta \\
&= \int_0^{\frac{\pi}{2}} [(1 + \cos\theta)^2 - 1] d\theta \\
&= \int_0^{\frac{\pi}{2}} (1 + 2\cos\theta + \cos^2\theta - 1) d\theta \\
&= \int_0^{\frac{\pi}{2}} (2\cos\theta + \cos^2\theta) d\theta \\
&= \int_0^{\frac{\pi}{2}} \left(2\cos\theta + \frac{1 + \cos 2\theta}{2} \right) d\theta \\
&= \left(2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right)_0^{\frac{\pi}{2}} \\
&= \left(2 + \frac{\pi}{4} \right)
\end{aligned}$$

Examples:

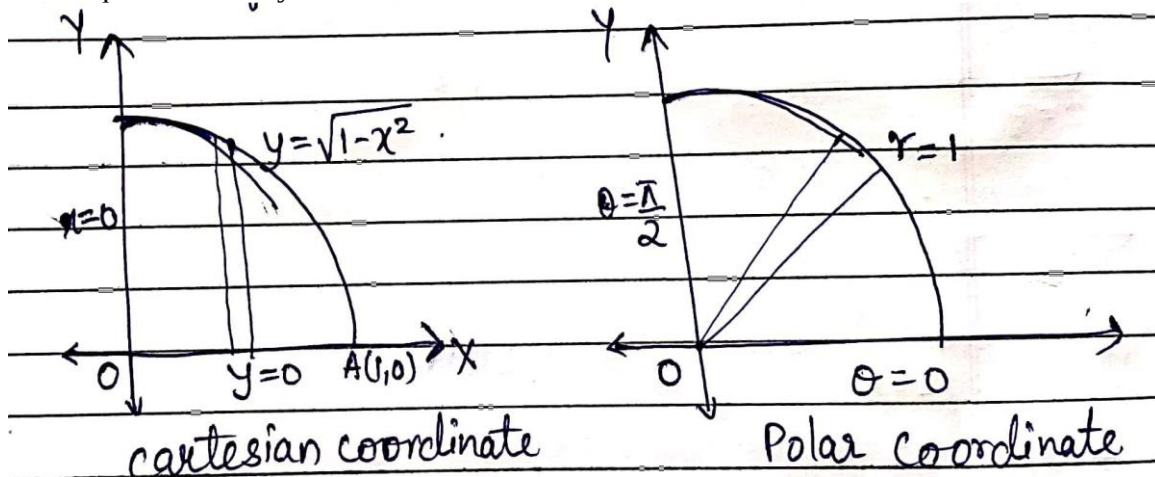
Example -1 Find the area lying inside the circle $r = a \sin \theta$ and outside the cardioids $r = a(1 - \cos \theta)$.

Example -2 Evaluate $\iint_R r \sqrt{a^2 - r^2} dr d\theta$, over the upper half of the circle $r = a \cos \theta$.

Change into Polar Form:

Example:1 Change the cartesian integral $\int_0^1 \int_0^{\sqrt{1-x^2}} e^{-(x^2+y^2)} dy dx$ into an equivalent polar integral.

Polar equation of $x^2+y^2=1$ is $r^2=1 \therefore r=1$



r varies from $r=0$ to $r=1$

θ varies from $\theta=0$ to $\theta=\frac{\pi}{2}$

$x = r \cos \theta$, $y = r \sin \theta \therefore x^2 + y^2 = r^2$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} e^{-(x^2+y^2)} dy dx = \int_0^{\pi/2} \int_0^1 e^{-(r^2)} r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^1 \frac{e^{-t}}{2} dt d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [-e^{-t}]_0^1 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [-e^{-1} + e^0] d\theta$$

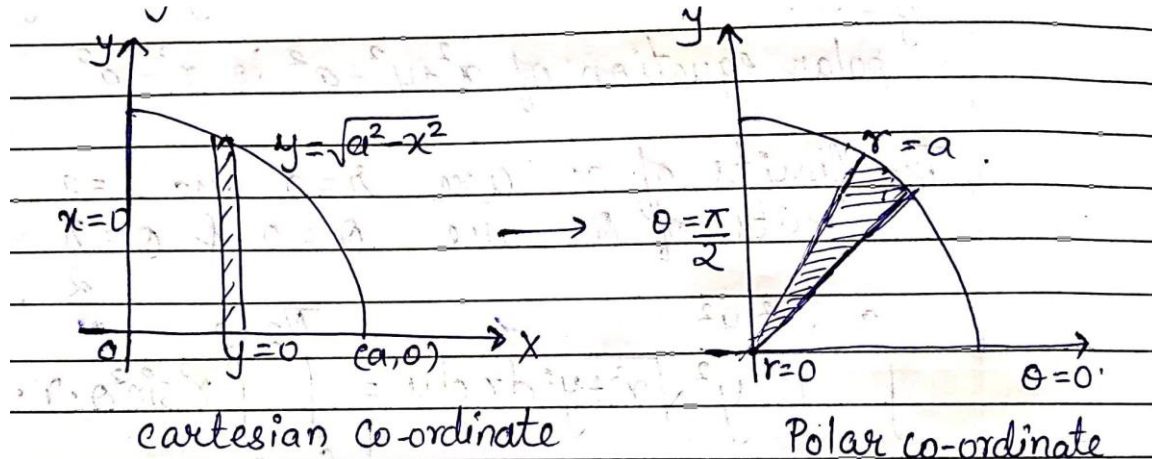
$$= \frac{-e^{-1} + 1}{2} [\theta]_0^{\pi/2}$$

$$= \frac{-e^{-1} + 1}{2} \times \frac{\pi}{2}$$

$$= \frac{(e-1)\pi}{4e}$$

EXAMPLE 2: Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) dy dx$ by changing into polar co-ordinates.

Solution :



$$x = r \cos \theta \text{ and } y = r \sin \theta \therefore dx dy = r dr d\theta$$

Given limits are $y = 0$ to $y = \sqrt{a^2 - x^2}$ and $x = 0$ to $x = a$.

$$y = \sqrt{a^2 - x^2}$$

$$\text{Then } x^2 + y^2 = a^2$$

Polar equation of given circle $x^2 + y^2 = a^2$ is $r^2 = a^2 \therefore r = a$;

Polar equation of the line $x = 0$ is $r \cos \theta = 0$ then $\theta = \frac{\pi}{2}$,

Polar equation of the line $y = 0$ is $r \sin \theta = 0$ then $\theta = 0$,

New limits for $r = 0$ to $r = a$, then $\theta = 0$ to $\theta = \frac{\pi}{2}$.

$$\begin{aligned} \int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) dy dx &= \int_0^{\frac{\pi}{2}} \int_0^a r^2 r dr d\theta \\ &= \int_0^{\frac{\pi}{2}} \left[\frac{r^4}{4} \right]_0^a d\theta \\ &= \frac{a^4}{4} [\theta]_0^{\frac{\pi}{2}} = \frac{\pi a^4}{8} . \end{aligned}$$

Example 3: Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2 + y^2} dx dy$ by changing into polar coordinate.

Solution: Given limits $x = 0$ to $x = \sqrt{a^2 - y^2}$ and $y = 0$ to $y = a$.

Polar equation of given circle $x^2 + y^2 = a^2$ is $r^2 = a^2 \therefore r = a$;

limits for $r = 0$ to $r = a$, then $\theta = 0$ to $\theta = \frac{\pi}{2}$.

$$\begin{aligned}
\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2+y^2} dx dy &= \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin^2 \theta r \cdot r \cdot dr d\theta \\
&= \int_0^{\frac{\pi}{2}} \int_0^a r^4 \sin^2 \theta dr d\theta \\
&= \int_0^{\frac{\pi}{2}} \left[\frac{r^5}{5} \right]_0^a \sin^2 \theta d\theta \\
&= \frac{a^5}{5} \int_0^{\frac{\pi}{2}} \left(\frac{1-\cos 2\theta}{2} \right) d\theta \\
&= \frac{a^5}{10} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\
&= \frac{a^5}{10} \left(\frac{\pi}{2} - 0 \right) \\
&= \frac{\pi a^5}{20}
\end{aligned}$$

Example:

- 1) Evaluate $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} x^2 dy dx$
- 2) Evaluate $\int_0^{2a} \int_0^{\sqrt{2x-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$
- 3) Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2+y^2) dx dy$

Triple Integrations

Evaluate the following triple integrations:

$$\begin{aligned}
1. \quad \int_1^2 \int_2^3 \int_0^1 xyz dx dy dz &= \int_1^2 \left[\int_2^3 \left[\int_0^1 xyz dz \right] dx \right] dy \\
&= \int_1^2 \left[\int_2^3 xy \left(\frac{z^2}{2} \right)_0^1 dx \right] dy \\
&= \frac{1}{2} \int_1^2 \left[\int_2^3 xy dx \right] dy \\
&= \frac{1}{2} \int_1^2 y \left(\frac{x^2}{2} \right)_2^3 dy = \frac{5}{4} \int_1^2 y dy = \frac{5}{4} \left(\frac{y^2}{2} \right)_1^2 = \frac{15}{8}
\end{aligned}$$

$$2. \quad \int_0^1 \int_0^{2-x} \int_0^{x-y} dz dy dx$$

$$\begin{aligned}
&= \int_0^1 \int_1^{2-x} [z]_0^{x-y} dy dx = \int_0^1 \int_1^{2-x} [x-y] dy dx = \int_0^1 \left[xy - \frac{y^2}{2} \right]_0^{2-x} dx = \int_0^1 \left[x(2-x) - \frac{(2-x)^2}{2} \right] dx \\
&= -\frac{1}{2} \int_0^1 [3x^2 - 8x + 4] dx = -\frac{1}{2} [x^3 - 4x^2 + 4x]_0^1 = -\frac{1}{2}
\end{aligned}$$

3. Evaluation of triple integrals yields a volume instead of an area.

Parallelepipeds (boxes) are summed rather than rectangles

$$\begin{aligned}
\int_1^2 \int_x^{2x} \int_0^{y-x} dz dy dx &= \int_1^2 \int_x^{2x} z \Big|_0^{y-x} dy dx = \int_1^2 \int_x^{2x} (y-x) dy dx = \int_1^2 \left[\frac{y^2}{2} - xy \right]_x^{2x} dx \\
&= \int_1^2 \left[\left(\frac{(2x)^2}{2} - x \cdot 2x \right) - \left(\frac{x^2}{2} - x \cdot x \right) \right] dx = \int_1^2 \frac{x^2}{2} dx \\
&= \frac{x^3}{6} \Big|_1^2 = \frac{8}{6} - \frac{1}{6} = \frac{7}{6}
\end{aligned}$$

Examples.

$$1 \int_0^1 \int_0^2 \int_0^{x+2} 6xz dx dy dz$$

$$2 \int_0^1 \int_0^{2x} \int_0^y 2xyz dx dy dz$$

$$3 \int_0^3 \int_0^1 \int_0^{\sqrt{1-z^2}} ze^y dx dy dz$$

$$4 \int_0^1 \int_0^z \int_0^y ze^{-y^2} dx dy dz$$

$$5 \int_0^1 \int_0^y \int_0^x \cos(x+y+z) dx dy dz$$